

HDOC Day-38

Date: 01-10-2026

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

Algorithm

1. Display the menu repeatedly until Exit is selected.
2. Read the user choice.
3. For choice 1, read a string and count spaces, digits, and special characters separately.
4. For choice 2, read a string and a character, then count its occurrences.
5. For choice 3, read a string and toggle the case of every alphabetic character.
6. For choice 4, terminate the program; otherwise display an invalid-choice message.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void countCharacters(char str[]) {
    int spaces = 0, digits = 0, special = 0, i;
    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] == ' ') spaces++;
        else if (isdigit((unsigned char)str[i])) digits++;
        else if (!isalpha((unsigned char)str[i])) special++;
    }
    printf("Spaces = %d\nDigits = %d\nSpecial characters = %d\n", spaces, digits, special);
}

void frequency(char str[], char ch) {
    int count = 0, i;
    for (i = 0; str[i] != '\0'; i++) if (str[i] == ch) count++;
    printf("Frequency of '%c' = %d\n", ch, count);
}

void toggleCase(char str[]) {
    int i;
    for (i = 0; str[i] != '\0'; i++) {
        if (islower((unsigned char)str[i])) str[i] = toupper((unsigned char)str[i]);
        else if (isupper((unsigned char)str[i])) str[i] = tolower((unsigned char)str[i]);
    }
    printf("Toggled string: %s\n", str);
}

int main() {
    int choice; char str[200], ch;
    do {
        printf("\n1. Count spaces, digits and special characters\n");
        printf("\n2. Count frequency of a character\n3. Toggle case of string\n4. Exit\nEnter choice: ");
        scanf("%d", &choice); getchar();
        switch (choice) {
            case 1:
                printf("Enter string: "); fgets(str, sizeof(str), stdin); str[strcspn(str, "\n")] = '\0';
                countCharacters(str); break;
            case 2:
                printf("Enter string: "); fgets(str, sizeof(str), stdin); str[strcspn(str, "\n")] = '\0';
                printf("Enter character: "); scanf(" %c", &ch); getchar(); frequency(str, ch); break;
            case 3:
                printf("Enter string: "); fgets(str, sizeof(str), stdin); str[strcspn(str, "\n")] = '\0';
                toggleCase(str); break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid choice!\n");
        }
    } while (choice != 4);
    return 0;
}
```

Kali Linux Terminal - Compilation and Sample Runs

```
(kali@kali)-[~/Desktop/cprog]
$ vi day38.c
(kali@kali)-[~/Desktop/cprog]
$ cc day38.c
(kali@kali)-[~/Desktop/cprog]
$ ./a.out

1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of string
4. Exit
Enter choice: 1
Enter string: hello
Spaces = 0
Digits = 0
Special characters = 0
```

```
(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter choice: 2
Enter string: hello
Enter character: l
Frequency of 'l' = 2

Enter choice: 3
Enter string: hello
Toggled string: HELLO

Enter choice: 4
Exiting program...
```

Test Cases

Case	Choice / Input	Expected Output
1	1; hello	Spaces = 0, Digits = 0, Special = 0
2	2; hello; l	Frequency of 'l' = 2
3	3; hello	HELLO
4	1; hello 2!	Spaces = 1, Digits = 1, Special = 1
5	2; hello; o	Frequency of 'o' = 1
6	4	Exiting program...

Result

The menu-driven C program was implemented successfully using separate functions for counting spaces/digits/special characters, finding character frequency, toggling case, and exiting the program.